

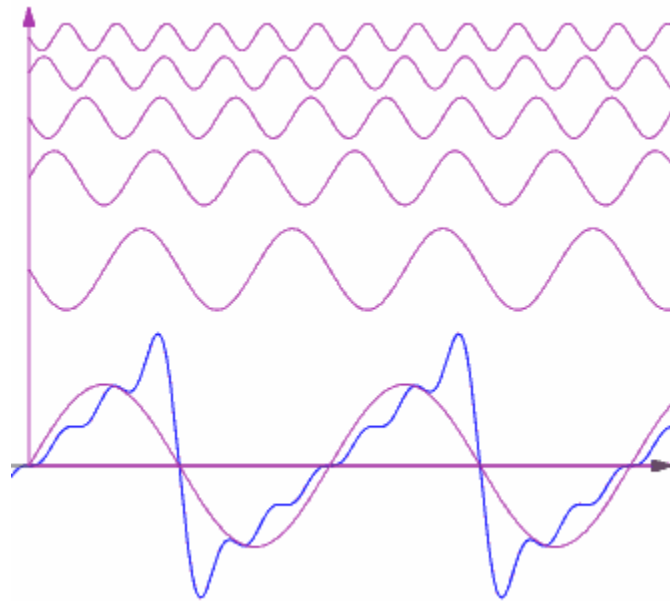
What Is a Filter?

A filter is a circuit capable of passing (or amplifying) certain frequencies while attenuating other frequencies. Thus, a filter can extract important frequencies from signals that also contain undesirable or irrelevant frequencies.

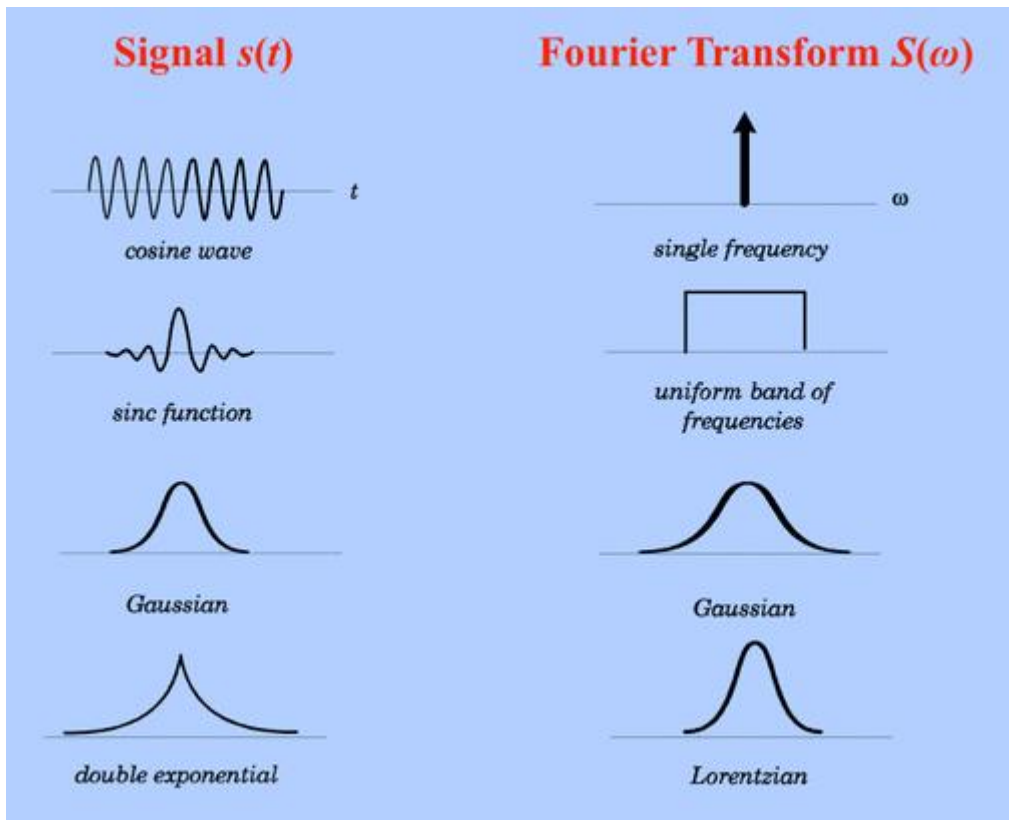
Filters can be placed in one of two categories: *passive* or *active*.

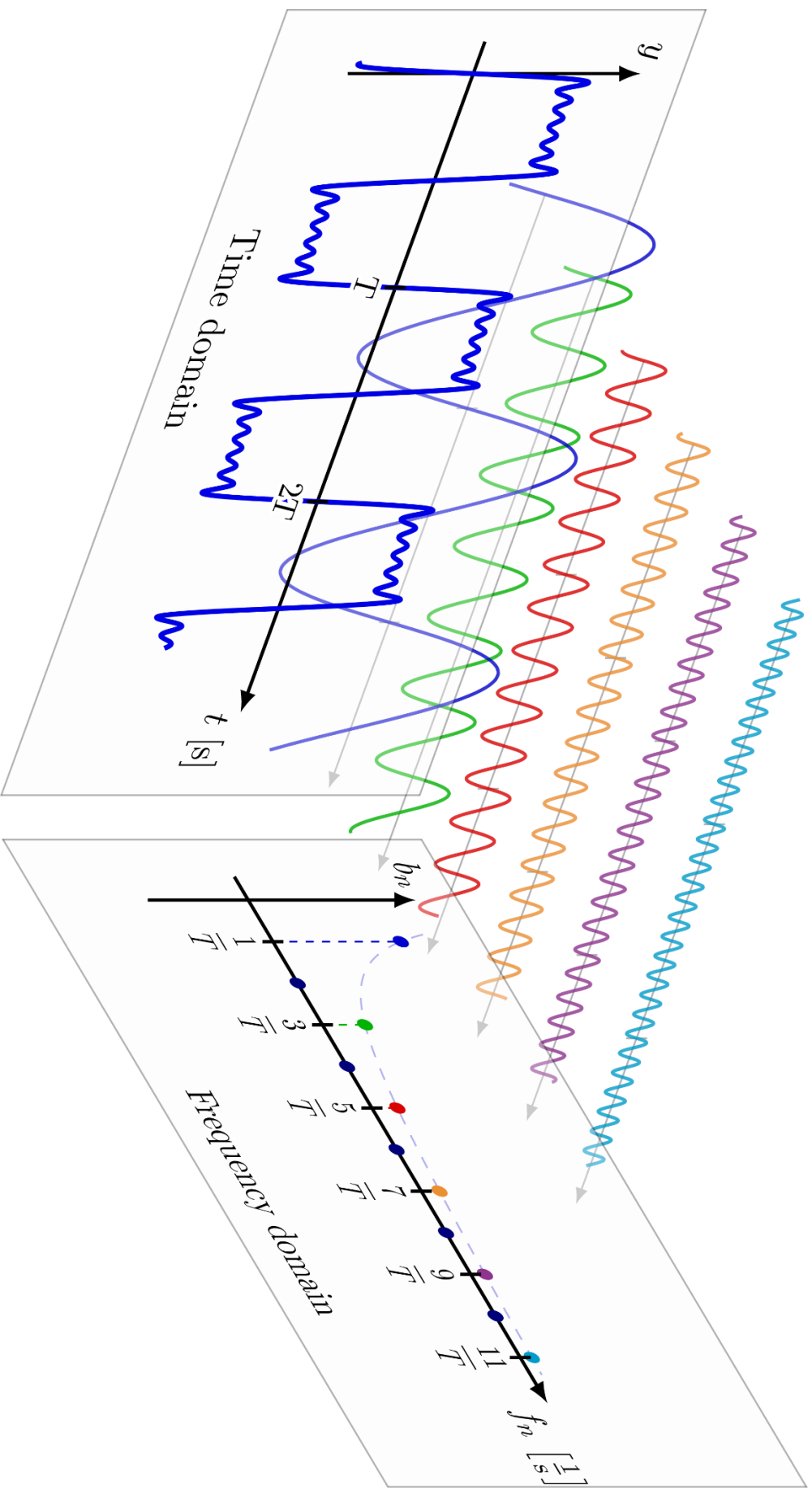
Passive filters include only passive components—[resistors](#), [capacitors](#), and [inductors](#). These components **do not** require an external power source to function and **cannot amplify signals**. In contrast, [active filters](#) use active components, such as [op-amps](#), in addition to resistors and capacitors, but not inductors. Active components **require an external power source** to operate and can **amplify signals** or perform switching operations.

Fourier Series

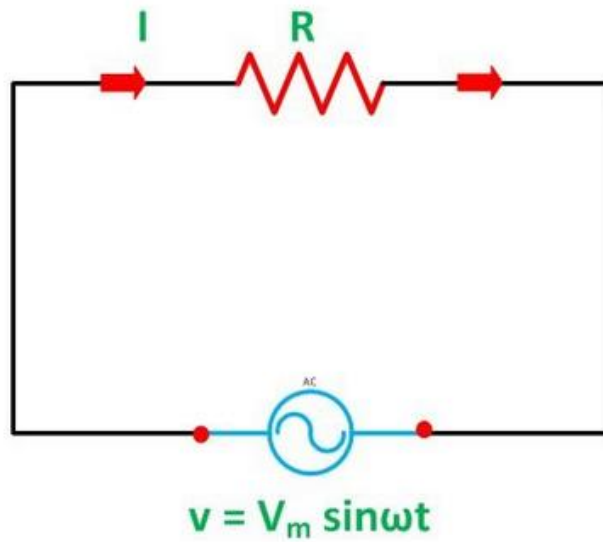


Fourier Transform





AC Circuits

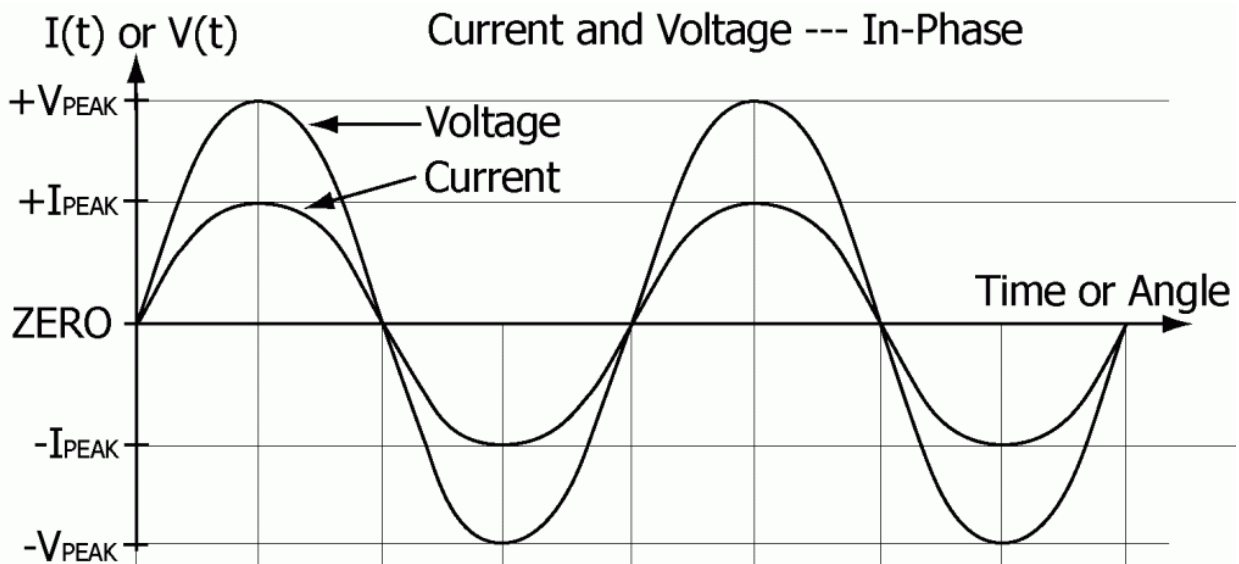


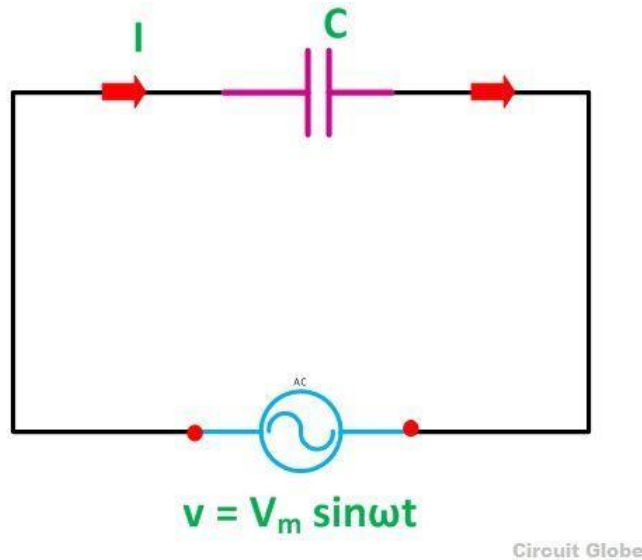
V_m : Max Voltage of source

ω : $2\pi f$ (oscillation/sec $\rightarrow$ rads/sec) Why? Mathematical Convenience, Direct Connection to Rotational Motion, Easier Representation of Phase Shifts

T : time

$$I = \frac{V_m \sin(\omega t)}{R} = I \sin(\omega t) \text{ (in-sync, same phase)}$$





$$V_c = CQ$$

$$I = dQ/dt$$

$$I = I_m \sin(\omega t)$$

then integrating both sides:

$$V_c = \frac{I}{C\omega} \sin(\omega t - 90^\circ)$$

This means that **voltage lags current by 90° in a purely capacitive circuit.**

Example: A **10 μF** capacitor is connected to a **100 V (rms), 50 Hz** AC source.

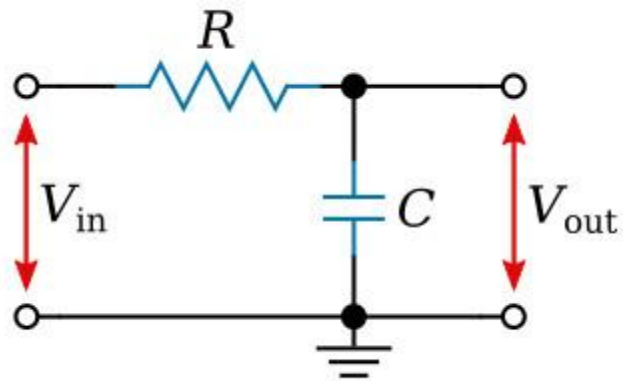
The opposition to AC current in a capacitor is called **capacitive reactance** X_c , given by:

$$X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

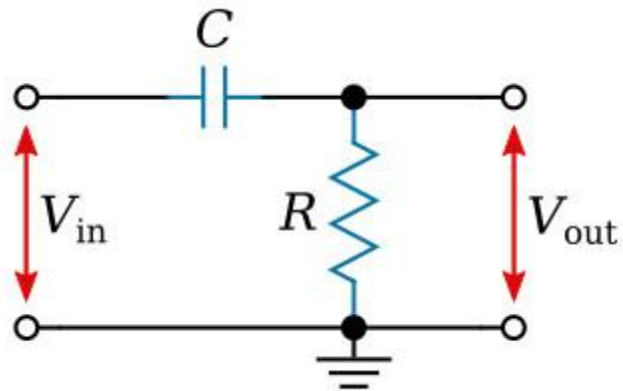
$$X_c \approx 31.8 \Omega$$

The capacitor behaves like a **31.8 Ω resistor**, but with a **90° phase shift** between voltage and current.

The capacitive reactance is dependent on frequency.



Low pass filter



High pass filter

Impedance (Z) is the total opposition a circuit presents to **alternating current (AC)**. It is the combination of **resistance (R)** and **reactance (X)** and is measured in **ohms (Ω)**.

$$Z = R + jX$$

$$Z = R + j\omega L + \frac{1}{j\omega C} \quad (2)$$

$$= R + j\left(\omega L - \frac{1}{\omega C}\right)$$